

Data Visualization

Seeing Patterns Clearly

Section 5 — Matplotlib and Seaborn

BEGINNER · EDA · PYTHON



Tables Are Not Always Enough

Tables are useful, but charts help us quickly see:

Comparison

Which category is highest

Concentration

Where values cluster

Spread

Whether values are spread out or tight

Outliers

Whether unusual values exist

Trend

Whether a pattern rises or falls

📌 Memory line: Charts help our eyes find patterns faster.

Start With the Question

Before choosing a chart, ask yourself:

→ What question am I answering?

→ Which columns are involved?

→ Are the columns numeric or categorical?

→ Am I comparing, showing distribution, or showing relationship?

A chart without a question is decoration. A chart with a question is evidence.

Chart Selection Map

Match the chart to the question:

Question	Chart	Example
Compare categories	Bar chart	Average spend by membership
Count categories	Count plot	Customers by city
Show ordered trend	Line chart	Spend by signup month
See distribution	Histogram	Monthly spend distribution
Compare spread	Box plot	Spend by membership
Compare two numbers	Scatter plot	Visits vs. spend



Memory line: The question chooses the chart.

Chart Anatomy

A readable chart needs:

- Clear title
- X-axis and Y-axis labels
- Meaningful scale
- Useful color choice
- Legend only when needed
- Enough whitespace



Quick check: Can someone understand this chart in 10 seconds?



If the reader needs a long explanation just to read the chart, the chart needs improvement.

Column Types Matter

Categorical

City · Membership ·
Signup month

Numeric

Monthly spend · Visits
per month

Column Type	Good Chart
Category count	Count plot or bar chart
Numeric distribution	Histogram
Numeric by category	Bar chart or box plot
Numeric vs. numeric	Scatter plot

📌 Memory line: Chart choice depends on column type.



Bar Chart — Compare Categories

A bar chart compares values across categories.

Example Question

Which membership type has the highest average monthly spend?

Data Needed

Category: membership · Measure: average monthly spend

Observation

Gold membership customers have the highest average monthly spend in this dataset.

Reading a Bar Chart

- ☐ Identify the tallest or longest bar
- ☐ Compare category sizes
- ☐ Read the axis label carefully
- ☐ Check whether the measure is count, sum, average, or median
- ☐ Do not assume color means something unless there is a legend

 Memory line: Bar height shows the comparison.

Bar Chart Mistake: Count vs. Average

Question 1

How many customers are in each membership group?

→ Uses counts

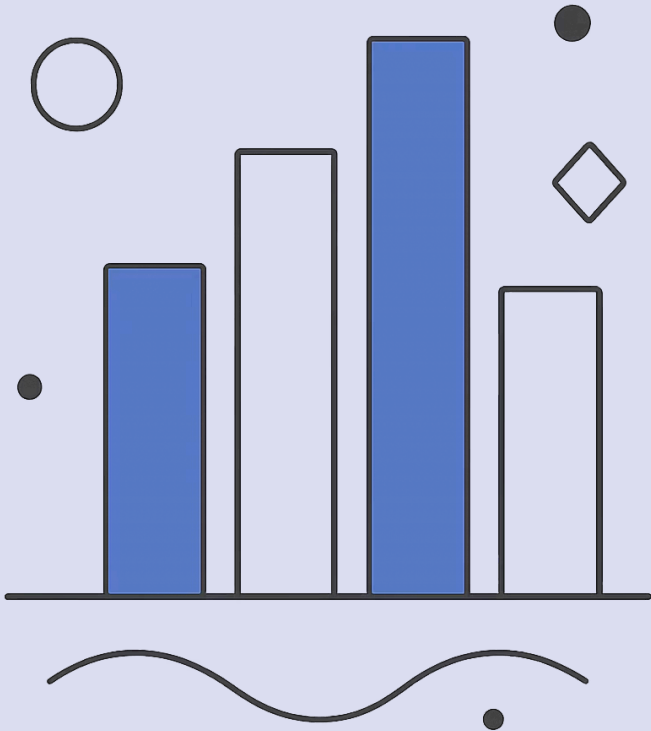
Question 2

What is the average monthly spend by membership group?

→ Uses averages



These are different questions. Always know what the y-axis is measuring.



Count Plot — Count Rows by Category

A count plot shows how many rows belong to each category.

City counts

Membership counts

Signup month counts

Product category counts



Memory line: Count plots answer: how many rows?

Reading a Count Plot

Interpretation

- Tallest bar has the most rows
- Shortest bar has the fewest rows
- Does not show average spend
- Does not show total revenue

Example observation: Mumbai and Delhi have the highest customer counts in this small dataset.



Customer count is not the same as customer value.

Line Chart — Show Ordered Change

A line chart shows change across an ordered sequence.

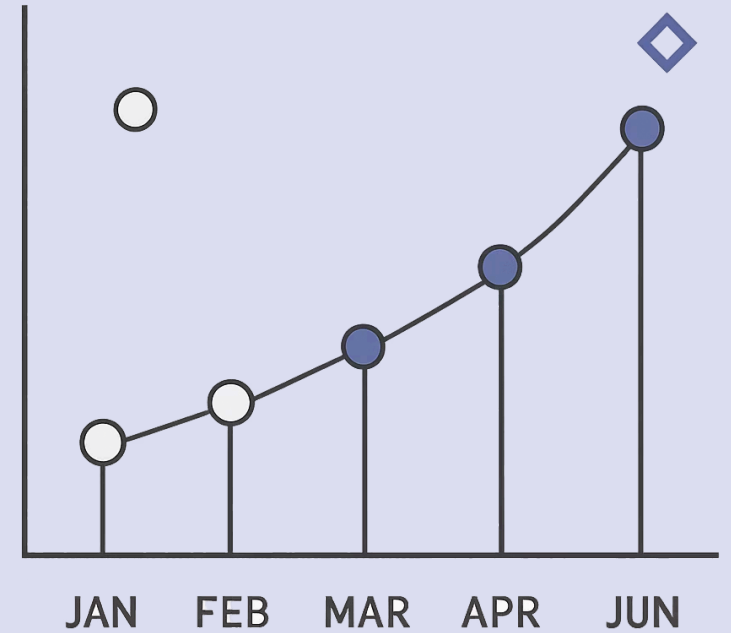
Good X-Axis Examples

Date · Month · Week number · Year · Ordered age group

Example Question

How does average monthly spend change across signup months?

📌 Memory line: Line charts need meaningful order.



Line Chart Caution

Do Not Connect Unordered Categories

Poor x-axis examples:

- City
- Product category
- Membership labels with no numeric order



The line suggests movement from one category to the next. If the x-axis has no real order, a line chart can mislead the audience.

Histogram — See Distribution

A histogram shows how values are distributed across ranges.

Low-value ranges

Middle-value ranges

High-value ranges

Concentration

Possible skewness

📄 Memory line: Histogram shows where numeric values fall.

Reading a Histogram

Key Points

- Each bar represents a value range (bin)
- Bar height shows how many values fall in that range
- Wider or narrower bins change the detail level
- Small datasets should be interpreted carefully

Example: Most monthly spend values appear concentrated around the middle ranges.



Do not read a histogram bar as one exact value — it represents a range.

Histogram Bin Size

Detail Level Matters

Few Bins

Simpler picture · Less detail · Easier to read at a glance

Many Bins

More detail · Can look noisy with small data · Reveals finer patterns

📌 Memory line: Bins change how the distribution appears.

Box Plot — Compare Spread

A box plot summarizes distribution and spread across groups.

Median

Middle line of the box

Middle 50%

The box itself (IQR)

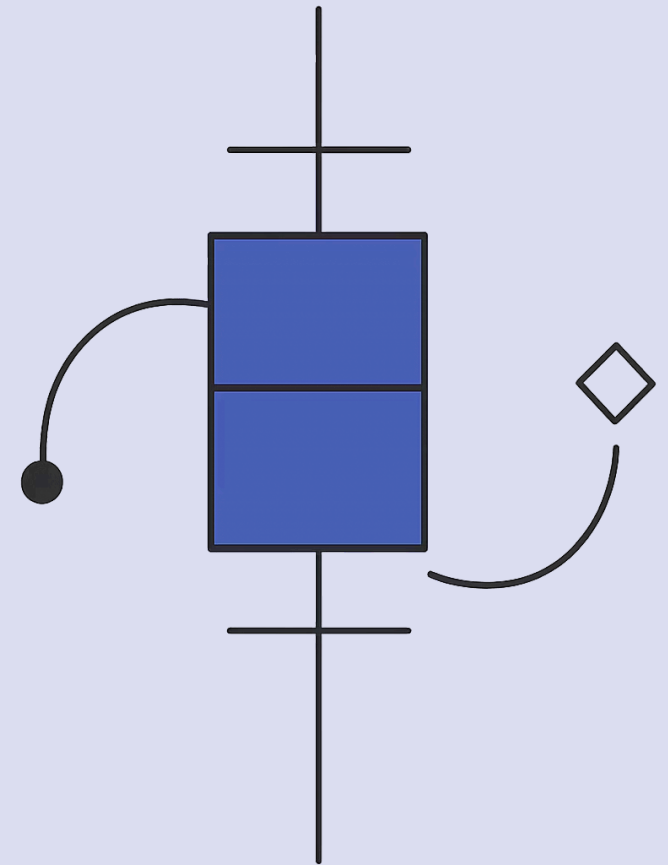
Whiskers

Typical low and high range

Outlier Points

Values outside the whiskers

📌 Memory line: The box shows the middle half of the data.



Box Plot Caution

A Possible Outlier Does Not Mean Wrong

A point outside the whiskers may be:

Real high-value customer

Rare but valid case

Data entry issue

Signal worth investigating

 Do not automatically delete it. A chart can flag a value for review, but it does not make the cleaning decision for us.

Scatter Plot — Compare Two Numeric Columns

A scatter plot compares two numeric variables. Each dot is one row.

Example Question

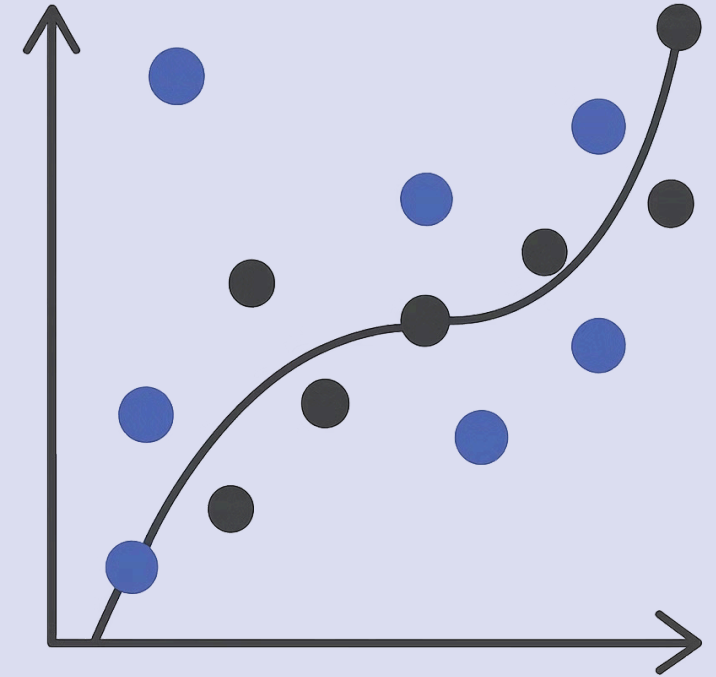
Do customers with more visits also tend to spend more?

X-Axis

Visits per month

Y-Axis

Monthly spend



Reading a Scatter Plot

Pattern	Meaning
Upward pattern	Same-direction movement
Downward pattern	Opposite-direction movement
Cloud with no direction	Weak or no clear relationship
Separate clusters	Groups may behave differently

Example observation: The scatter plot suggests visits and spending move together somewhat.

Scatter Plot Caution

Association Is Not Cause

What a scatter plot CAN show

More visits and higher spend appear together — an association.

What it CANNOT prove

More visits caused higher spend. Correlation does not establish causation.

📌 Memory line: Scatter plots show relationships, not proof of cause.

Use Color With Meaning

Good Color Use

- Highlight one key category
- Separate groups clearly
- Keep one color for one meaning

Poor Color Use

- Too many colors
- Random colors with no meaning
- Color without a legend
- Color used only for decoration

 Color should reduce confusion, not add confusion.

Chart Titles

State the Question

1

Weak
Chart 1

2

Better
Average Monthly Spend by Membership

3

Strong
Gold Customers Have the Highest Average Monthly
Spend

📌 Memory line: A title should help the reader know what to look for.

Axis Labels

Say What Is Measured

Axis labels should answer:

- What is on the x-axis?
- What is on the y-axis?
- Is the measure count, average, total, or percentage?
- What units are used?



Example:

X-axis: Membership

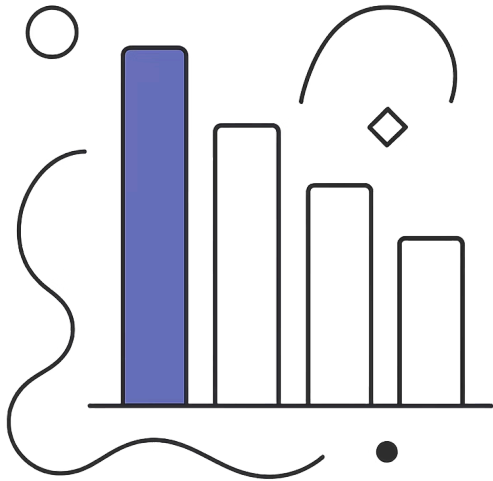
Y-axis: Average Monthly Spend (INR)



Unlabeled axes force the audience to guess.

Sorting Bars

Make Comparisons Easier



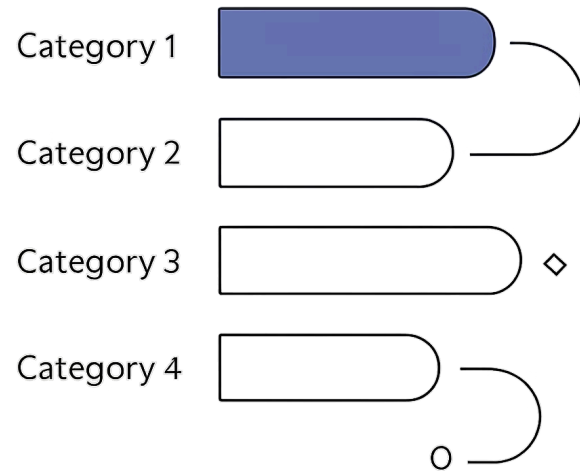
Sort bars when order helps comparison:

- Highest to lowest average spend
- Largest to smallest customer count
- Lowest to highest score

 Sorted bars make the main pattern easier to see and reduce visual searching.

Horizontal Bar Charts

Useful for Long Labels



Use horizontal bars when:

- Category names are long
- Many categories exist
- You want easy label reading

Example: Customer Count by City

❏ Use horizontal bars when vertical labels become crowded.

Annotating Charts

Highlight the Main Evidence

Annotation means adding a small label to guide attention. Useful for:



Highest Value

Draw attention to the peak



Lowest Value

Flag the minimum



Key Threshold

Mark an important benchmark



Possible Outlier

Signal a value worth reviewing

 Memory line: Annotation tells the reader what not to miss.

Pair Charts With Tables

Charts show the pattern

A chart makes the visual trend immediately clear.

Tables show exact values

Membership	Avg Spend
Gold	1417
Silver	992
Bronze	863



Charts and tables support each other. Charts show pattern; tables show exact evidence.

Misleading Charts

Technically Correct Can Still Mislead

Missing title or labels

Y-axis scale exaggerates differences

Too many colors

Wrong chart type

Line chart for unordered categories

Small dataset shown as a big conclusion

📌 Memory line: A chart should clarify, not manipulate.

Small Dataset Caution

Do Not Overread the Picture

Small datasets are useful for learning. But:

- One row can change the chart
- Group comparisons may be unstable
- Histogram shapes may be noisy
- Scatter patterns may be weak evidence



Use charts to learn the workflow. Use larger validated data for serious business decisions.



Matplotlib — Chart Control

Matplotlib is a Python library for creating charts with detailed control.

Titles & Labels

`plt.title()` · `plt.xlabel()` · `plt.ylabel()`

Display & Size

`plt.show()` · `plt.figure(figsize=...)`

Styling

Colors · Custom formatting · Figure size



Memory line: Matplotlib gives detailed chart control.

Seaborn — Statistical Charts Made Easier

Seaborn is built on top of Matplotlib. It creates common statistical charts with less code.

Box plots

```
sns.boxplot()
```

Scatter plots

```
sns.scatterplot()
```

Count plots

```
sns.countplot()
```

Grouped visuals

Cleaner default styling

Writing Chart Observations

Turn Visual Evidence Into Meaning

1

Weak

The chart is good.

2

Better

Gold membership customers have the highest average monthly spend.

3

Stronger

Gold membership customers have the highest average monthly spend, but the dataset is small, so this should be validated with more rows.



Memory line: A chart observation should include evidence and meaning.

Chart Observation Pattern

Evidence + Meaning + Caution

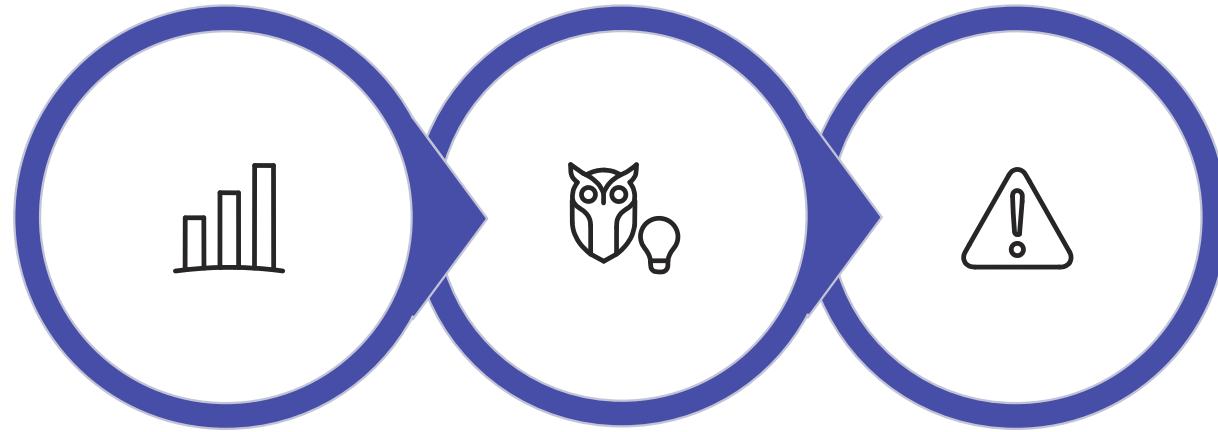


Chart
Evidence

Plain Meaning

Caution /
Check

The tallest bar is Gold, so Gold has the highest average monthly spend.

The histogram shows spend values concentrated in the middle range.

The scatter plot suggests visits and spend move together, but it does not prove cause.

Notebook 05 — Lab Mapping

In Notebook 05, learners will practice:

Chart Selection

Bar, Line, Histogram

Box & Scatter Plots

Count Plots

Formatting & Annotation

Writing Observations

 Notebook 05 turns visual thinking into Matplotlib and Seaborn practice.

Trainer Demo Flow

Recommended Live Sequence

01

Start with a question

02

Choose the columns

03

Choose the chart type

04

Create the chart

05

Add title and labels

06

Read the main pattern

07

Compare with a table if useful

08

Write one observation with caution if needed

📌 Memory line: Question first. Chart second. Observation third.

Discussion Prompt

Choose the Chart

Question 1

Which chart would you use to compare average spend by membership?

Expected answer: Bar chart

Question 2

Which chart would you use to see the distribution of monthly spend?

Expected answer: Histogram

Section Summary

Charts answer questions

Column type drives chart choice

Bar charts compare categories

Count plots count rows

Line charts need ordered x-axes

Histograms show distributions

Box plots show spread

Scatter plots show relationships

Titles, labels, color & scale matter

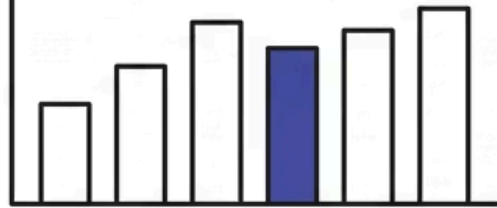
Observations need evidence & meaning

📌 Memory line: Good visualization makes data easier to understand and explain.

Quick Reference

Chart Selection at a Glance

Bar Chart



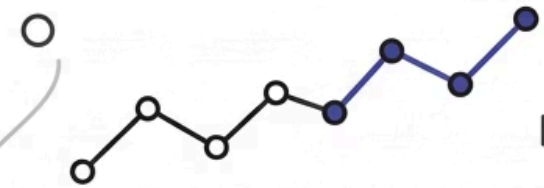
Bar Chart: Comparing category values, like ave. spend by member type.

Count Plot



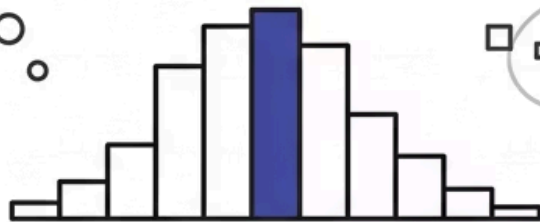
Count Plot: Counting data rows per category, e.g., customers by city.

Line Chart



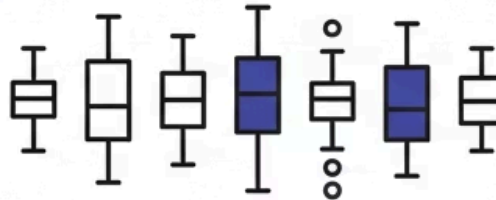
Line Chart: Tracking values in order, e.g., spend by signup month.

HISTOGRAM



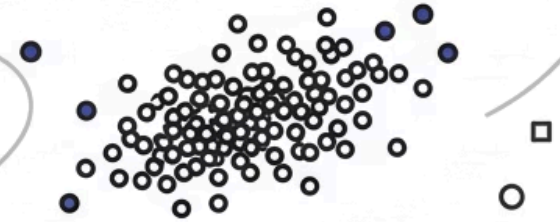
Histogram: Showing the spread of numeric values, like monthly spend range

BOX PLOT



Box Plot: Showing spread & outliers in data, e.g., spend by membership group

SCATTER PLOT



Scatter Plot: Finding relationships between two numbers, e.g., visits vs spend

Data Visualization

Seeing Patterns Clearly

Question First

Every chart starts with a question

Right Chart

Column type drives the choice

Read Responsibly

Evidence + meaning + caution

SECTION 5 COMPLETE

